The Bombay Salesian Society's

DON BOSCO INSTITUTE OF TECHNOLOGY

Premier Automobiles Road, Kurla West, Mumbai - 400070

IEEE- DBIT

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Report on – COLOHACKS (HARDWARE)

Title: Colohacks (Hackathon - Hardware)
Date: 28th & 29th March, 2026 (overnight 24hrs)
Time: 10 a.m. - 4 p.m.
Venue: B - WING 1ST FLOOR EXTC DEPARTMENT

Target Audience: Open For All

No. of Participants Present: 92

No. of Girl Participants Present: 21

No. of Boy Participants Present: 71

Mentor (Alumnus 2025): Dibyarupa Pradhan, Prem Singh, Mayuri Kadam, and Rutvik Patil

Judges Panel: Mr. Akhil Pathak, Mr. Jitendra A. Chaurasiya, Ms. Sayli Hareshwar Tandel, Ms. Rohini Mhatre, Mr. Perez Samko Dasan, and Mr. Arbaz Khan

Faculty Coordinator: Ms Freda Carvalho

Objectives:

- Encourage innovation and practical problem-solving through hardware-based projects
- Provide a platform for students from multiple colleges to collaborate and compete
- Promote hands-on learning and technical skill development in real-time
- Foster teamwork, time management, and endurance in a high-pressure environment
- Enable participants to transform ideas into working prototypes
- Facilitate continuous mentorship and guidance to improve project quality
- Conduct a structured evaluation process through multiple judging rounds

Outcomes:

- Participants developed functional hardware prototypes within a limited time frame
- Improved technical, design, and implementation skills among students
- Enhanced ability to work efficiently under pressure and tight deadlines
- Strengthened team coordination and task distribution strategies
- Exposure to real-world problem-solving and innovation challenges
- Identification of top 10 high-performing teams through rigorous evaluation
- Boosted confidence, creativity, and competitive spirit among participants

Detailed Report:

The 24-hour Hardware Hackathon organised by IEEE DBIT under Colosseum 2026 was a highly intensive and innovation-driven event conducted from 10:00 AM on 24th March to 4:00 PM on 25th March, bringing together participants from multiple colleges to design and develop real-time hardware solutions addressing practical, real-world problems. The event began with a rigorous selection process conducted through Unstop, where teams were required to submit a detailed presentation in the form of a PPT along with a video explanation of their proposed ideas, out of which only 23 teams were shortlisted for the final offline hackathon. The hackathon officially commenced at 10:00 AM, with all selected teams beginning work simultaneously on their respective projects, showcasing immense dedication, creativity, and technical expertise. Participants worked continuously throughout the 24-hour duration, effectively managing time and energy by taking short breaks in shifts while ensuring consistent progress in development and implementation. Continuous mentorship was provided throughout the event, where experienced mentors guided teams in refining their concepts, troubleshooting technical challenges, and improving the feasibility and scalability of their solutions. The evaluation process was structured into three rounds of judging to ensure a fair and comprehensive assessment, with the first round focusing on idea validation and initial progress, the second round evaluating implementation and working prototypes, and the final round determining the top 10 teams based on innovation, execution, and real-world applicability. The top-performing teams presented highly impactful and diverse solutions across domains such as healthcare, smart cities, environmental sustainability, and AI-driven systems. Several healthcare-focused innovations stood out, including a portable IoT-enabled life-saving kit designed for emergency and rural healthcare scenarios, a distributed mini ICU system aimed at providing affordable and continuous patient monitoring beyond hospitals, and a smart rehabilitation glove using AI and servo-assisted mechanisms to support physiotherapy and stroke recovery. Additionally, an IoT-based smart inhaler was developed to monitor usage patterns, prevent overuse, and provide real-time tracking through cloud integration. In the domain of smart cities and environmental monitoring, teams proposed innovative solutions such as an environmental monitoring station capable of tracking multiple air quality parameters using IoT and long-range communication technologies, and a water quality monitoring system designed to provide real-time analysis of parameters like turbidity, pH, and TDS for safe drinking water. AI-based systems also played a significant role, with one team developing an adaptive traffic management system using computer vision and real-time analytics to optimise traffic flow and reduce congestion and accidents. Another notable innovation was a multimodal neuro-health monitoring system that integrates brain signals, heart activity, and environmental data to detect early neurological imbalances and provide deeper health insights. Overall, the hackathon successfully created a highly competitive and collaborative environment that encouraged innovation, critical thinking, and teamwork under pressure, while providing participants with valuable hands-on experience and exposure to real-world engineering challenges. The event concluded with the selection of the top 10 teams, marking it as a successful and impactful initiative that highlighted the potential of student-driven innovation in solving pressing societal issues.

Judges Panel and Alumni Interaction:

Distinguished Panel of Judges

The 24-hour Hardware Hackathon, **ColoHacks 2026**, organised by IEEE-DBIT under Colosseum 2026, was evaluated by a distinguished panel of six industry professionals possessing expertise in embedded systems, electronics, hardware design, robotics, automation, and product development. The judging panel comprised five alumni of Don Bosco Institute of Technology (DBIT), Mumbai, along with one external industry expert. Their participation brought valuable industry perspectives to the event and strengthened the connection between the institution, its alumni, and current students.

The judges evaluated participating teams through multiple rounds, focusing on innovation, technical implementation, prototype functionality, feasibility, and real-world applicability. Their professional experience helped provide a practical perspective on the solutions developed by participants.

1. Mr. Akhil Pathak

DBIT Alumnus - Batch of 2016-2017

Mr. Akhil Pathak, an alumnus of Don Bosco Institute of Technology, was invited as a member of the judging panel for ColoHacks 2026. His participation contributed to the institution's initiative of involving former students in technical events and encouraging interaction between alumni and current students.

2. Mr. Jitendra A. Chaurasiya

Assistant Manager, Tata Advanced Systems Ltd. - SED

DBIT Alumnus - Batch of 2014

Mr. Jitendra A. Chaurasiya is an experienced professional specialising in embedded systems, electronics design, and defence-related product development. He holds a Bachelor's degree in Electronics and Telecommunication Engineering from DBIT and an M.Tech in Embedded Systems from BITS Pilani. His expertise includes PCB design, FPGA-based systems, microcontrollers, electronic product development, hardware troubleshooting, and system-level testing. His extensive industrial experience contributed valuable technical perspectives to the evaluation of hardware-based innovations.

3. Ms. Sayli Hareshwar Tandel

Deputy Manager - Embedded Software Engineer, Lauritz Knudsen, a unit of Schneider Electric India Pvt. Ltd.

DBIT Alumna - Batch of 2019–2020

Ms. Sayli Tandel is an embedded software professional specialising in microcontroller programming, real-time embedded systems, hardware debugging, and industrial automation. Her expertise includes STM32, Renesas and NXP microcontrollers, communication protocols, RTOS, and embedded software development. She has also been actively involved in technical workshops and student activities during her academic journey. Her association with ColoHacks 2026 highlighted the continued involvement of DBIT alumni in promoting technical innovation and practical engineering skills.

4. Ms. Rohini Mhatre

Design Engineer, Selec Controls Pvt. Ltd.

DBIT Alumna – Batch of 2021–2022

Ms. Rohini Mhatre is a design engineer with professional experience in hardware and firmware development for Programmable Logic Controllers (PLCs). Her technical expertise includes embedded C programming, microcontrollers, communication protocols, circuit design, hardware testing, and industrial automation. She has worked on PLC development, power supply design, functional testing, and product validation in accordance with industry standards. Her participation added valuable expertise in embedded electronics, hardware reliability, and practical implementation.

5. Mr. Perez Samko Dasan

Electronic Engineer, PLA Electro Appliances Pvt. Ltd.

DBIT Alumnus – Batch of 2021–2022

Mr. Perez Samko Dasan is an electronics professional with experience in PCB development, electronic testing, calibration, and quality assurance. His professional work involves developing and testing electronic circuits used in radiation monitoring devices, including survey meters and contamination monitors. His technical skills include embedded systems, PCB designing, circuit debugging, and instrumentation. His involvement in the judging panel provided an industry-oriented perspective on hardware functionality, testing procedures, and product reliability.

6. Mr. Arbaz Khan

Design Engineer (Aerospace), Cyient Ltd.

External Industry Expert

Mr. Arbaz Khan holds an M.Tech in Advanced Manufacturing and Design from the Indian Institute of Technology, Jodhpur. He specialises in mechanical design, aerospace engineering, robotics, structural analysis, and simulation. His professional experience includes finite element analysis, CAD modelling, composite structures, and aerospace product design. He has also worked on robotics projects involving autonomous navigation, quadruped robots, and motion control. His multidisciplinary expertise contributed an additional perspective to the evaluation of innovative hardware projects.

Alumni Interaction and Industry-Academia Engagement

One of the significant highlights of ColoHacks 2026 was the involvement of **five distinguished alumni of Don Bosco Institute of Technology** as members of the judging panel. The participation of alumni from different graduating batches created an opportunity to strengthen alumni–student relationships and promote meaningful interaction between academia and industry.

Through the judging and evaluation sessions, participating students had the opportunity to present their innovative ideas, explain the technical implementation of their projects, demonstrate working prototypes, and respond to questions from experienced industry professionals. This provided a platform for professional interaction and exposed students to the practical considerations involved in engineering design, product development, testing, and real-world problem-solving.

The presence of alumni working in areas such as embedded systems, electronics design, industrial automation, defence technology, and hardware development also demonstrated the diverse professional opportunities available to engineering graduates. Their involvement helped bridge the gap between classroom learning and industry expectations while encouraging students to approach their projects with greater technical depth, creativity, and practical relevance.

The interaction further strengthened the institute's alumni network by providing former students with an opportunity to reconnect with their alma mater and contribute to the development of aspiring engineers. Such initiatives play an important role in encouraging professional networking, knowledge exchange, career awareness, and continuous alumni engagement.

Overall, the distinguished judging panel and active alumni participation significantly enhanced the academic and professional value of ColoHacks 2026, making the event not only a platform for technical competition but also an opportunity for industry exposure, mentorship, and meaningful alumni–student interaction.

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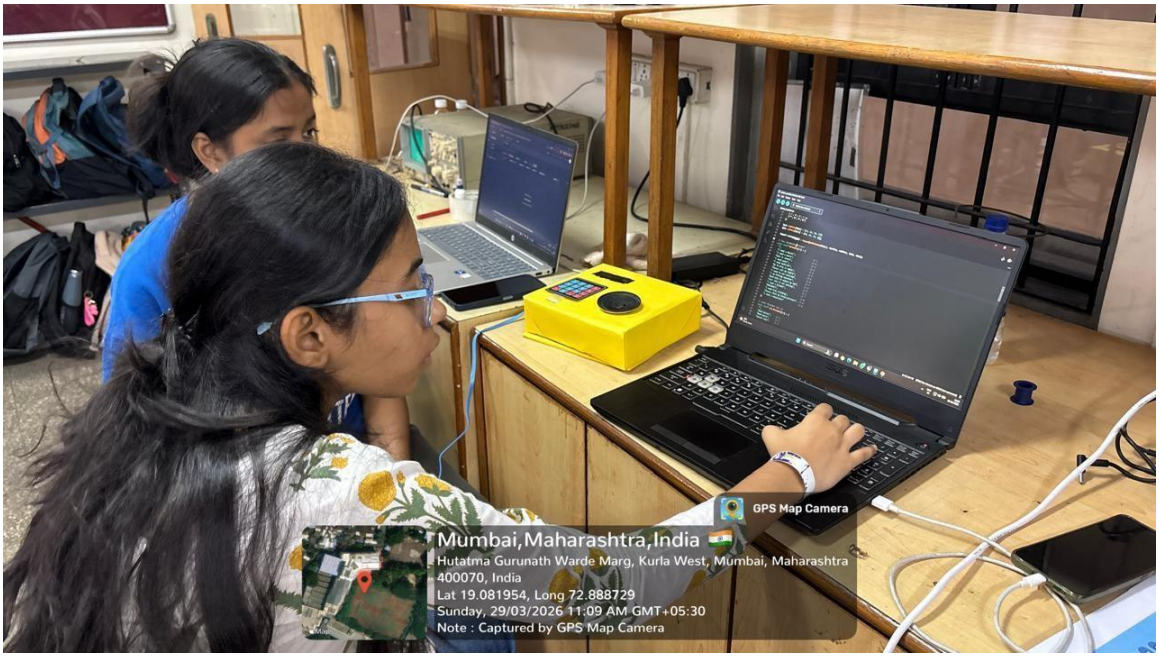
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Snapshot of the Event:



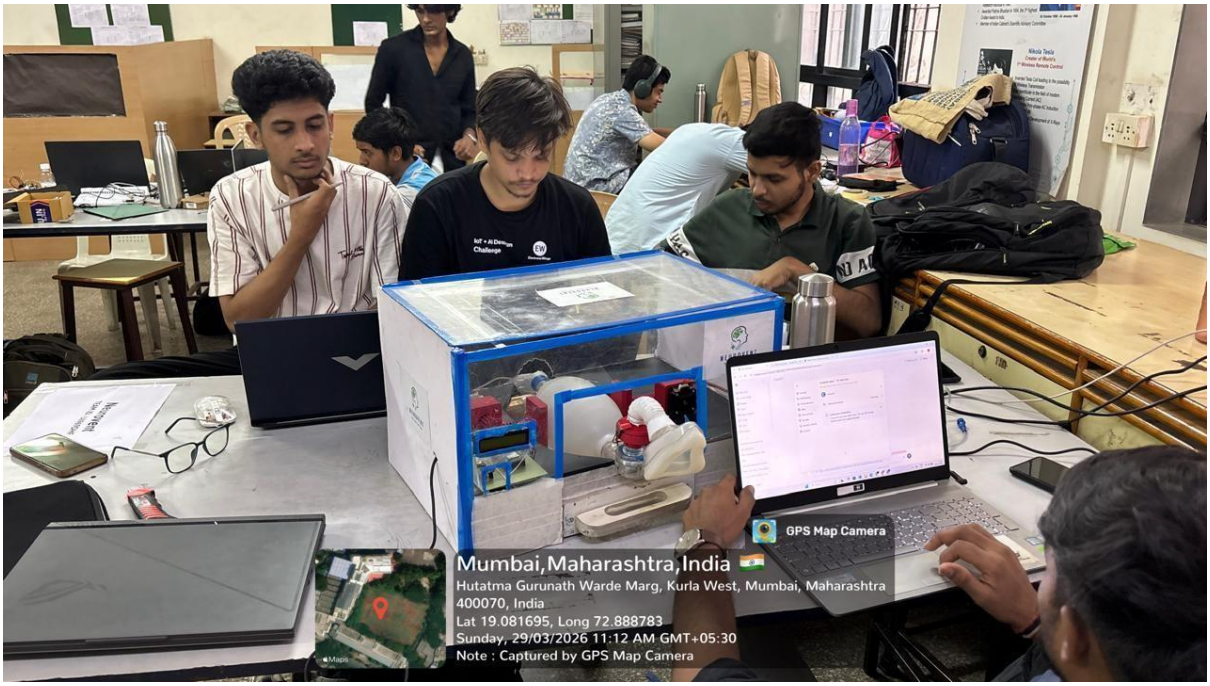


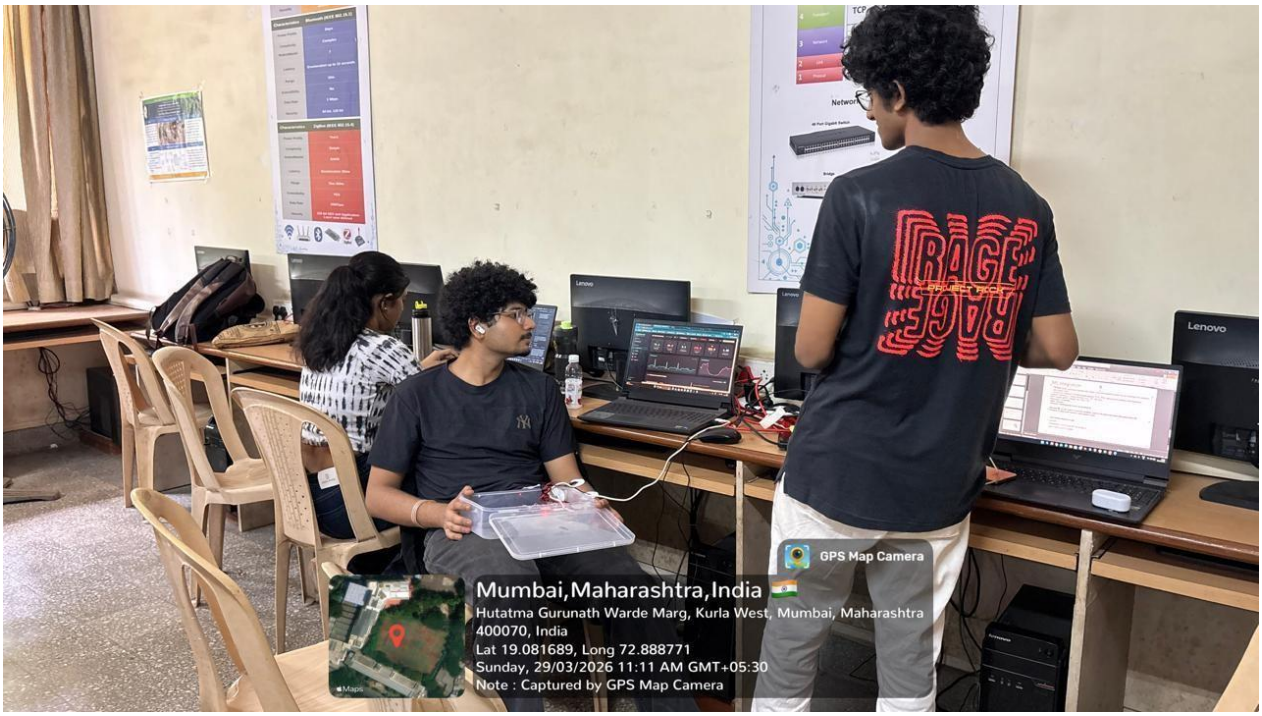
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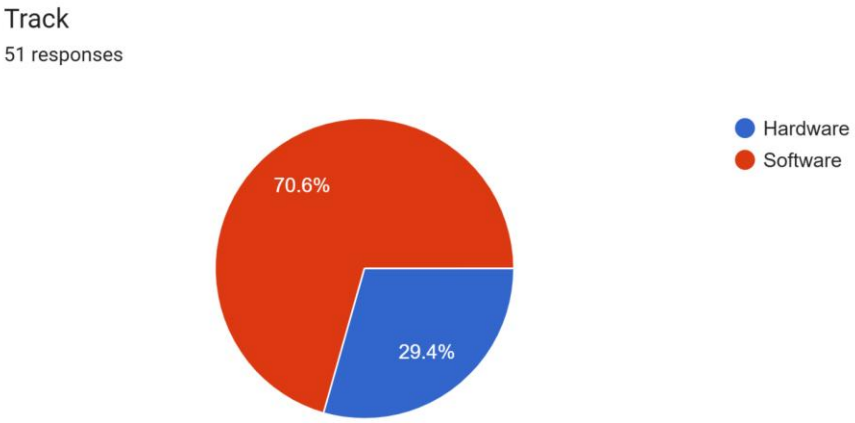
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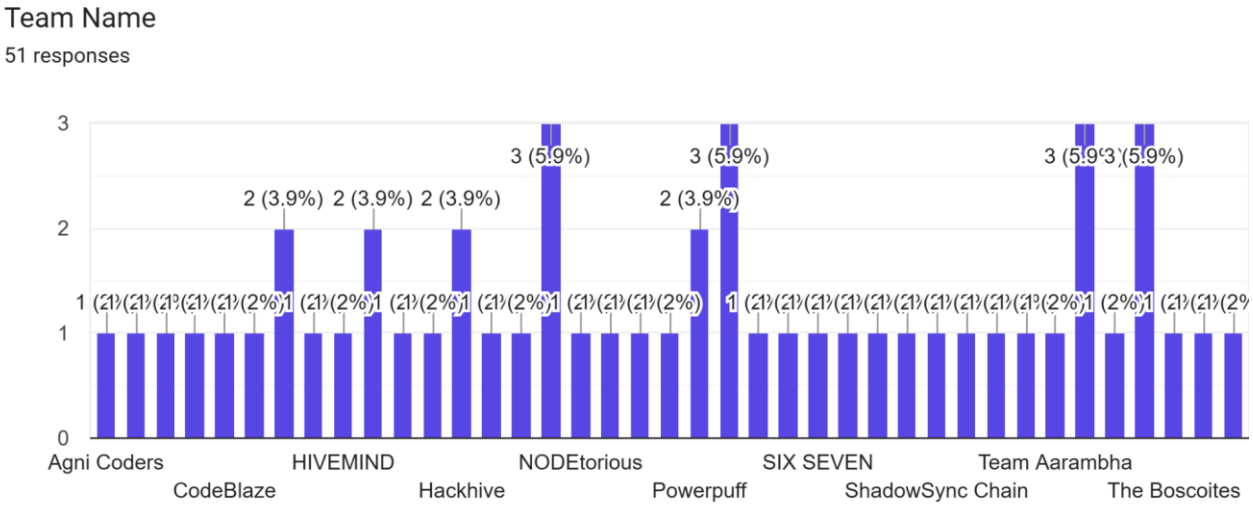


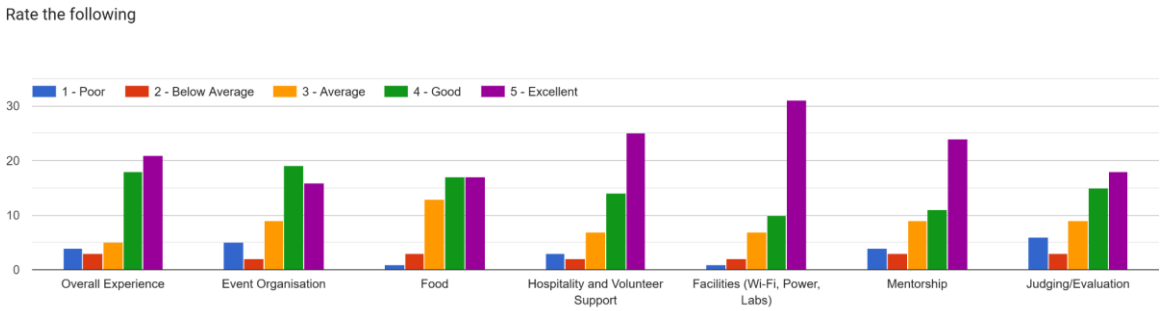


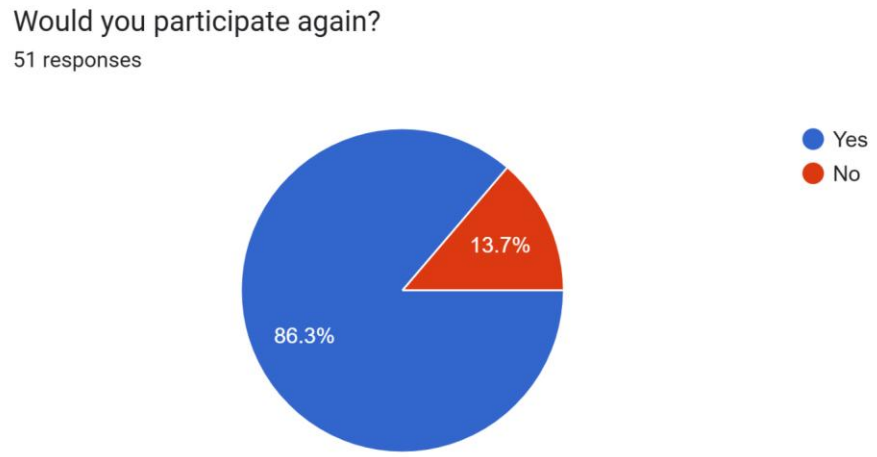

GPS Map Camera
Mumbai,Maharashtra,India
Hutatma Gurunath Warde Marg, Kurla West, Mumbai, Maharashtra 400070, India
Lat 19.081689, Long 72.888771
Sunday, 29/03/2026 11:11 AM GMT+05:30
Note : Captured by GPS Map Camera

Feedback Analysis:









The feedback analysis collected from 51 participants reflects an overall positive response towards the event, highlighting its successful organisation and execution. The track-wise distribution indicates that 70.6% of participants belonged to the Software track, while 29.4% represented the Hardware track, demonstrating greater participation in software-based projects. The team-wise analysis shows involvement from various teams, reflecting diverse participation and engagement throughout the event. The ratings received for different aspects, including overall experience, event organisation, food arrangements, hospitality, volunteer support, facilities, mentorship, and judging and evaluation, were predominantly positive, with the majority of responses falling under the Good and Excellent categories. Facilities such as Wi-Fi, power supply, and laboratory arrangements received particularly high appreciation, indicating effective infrastructure management. Hospitality, volunteer coordination, and mentorship were also well received, highlighting the efforts of the organising committee in ensuring a smooth and supportive experience for participants. However, certain areas, particularly food arrangements and aspects of event coordination, showed scope for further improvement. Notably, 86.3% of respondents expressed their willingness to participate in future editions, while only 13.7% indicated otherwise, demonstrating a high level of participant satisfaction. Overall, the feedback suggests that the event successfully provided a collaborative, engaging, and technically enriching platform for participants to showcase their skills, exchange ideas, and gain practical experience. The insights gathered will help the organising team further enhance event quality, strengthen participant engagement, and ensure an even better experience in future editions.

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Event Poster:



Evaluation and Awards

The hackathon featured a prize pool worth ₹1,50,000, with the top three teams awarded ₹45,000, ₹25,000, and ₹15,000 respectively. In addition, domain-specific awards of ₹5,000 each were presented in the Software track for excellence in AI/ML, App Development, and Web Development.
Final presentations were conducted before a panel comprising five judges for the Software track and six judges for the Hardware track, ensuring a fair and comprehensive evaluation process.
Shortlisted teams were also recognized with certificates and hampers, acknowledging their efforts and innovative contributions.

Valedictory Ceremony

The closing ceremony of ColoHacks 2026 was held at 7:00 p.m. at Mondini Hall. The session commenced with a welcome address by Zubia Sarang, followed by remarks from Dr. Sudhakar Mande, who congratulated the top teams and appreciated the efforts of all participants and organizers.
Distinguished guests and judges were felicitated for their valuable contributions to the event. Winners across both Software and Hardware tracks were awarded prizes and certificates, recognizing their innovation, technical excellence, and problem-solving capabilities.
The ceremony concluded with a vote of thanks delivered by Kartik Dandelia, expressing gratitude to the faculty, organizing committee, mentors, and participants. The event marked a successful conclusion to an intense and inspiring hackathon experience.



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WINNERS OF COLOHACKS 2026 - HARDWARE

1st place- Tech Tribe

Rizvi College of Engineering, Mumbai (Harsh Sankhe, Sneha Kaida)



2nd place- RJMASK

Don Bosco Institute of Technology, Mumbai (Mrunmai Patekar, Blossom Mascarenhas, Raman Sharma)



3rd place- Mistreated Devs

Shah and Anchor Kutcchi Engineering College, Mumbai (Tanish Nagrekar, Manav Kothari, karanjawane parth, yaswanth)



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Social Media Links:

https://www.instagram.com/ieee_dbit/

https://www.instagram.com/colosseumdbit?igsh=NDN4bWJ6M2c3Zmtn&utm_source=qr

Registration Details:

In case the event is open for non-DBIT students:

No. of DBIT Students: 36

No. of non-DBIT students: 56

Sr. no.	Team Name	Candidate's Name	Specialization
1.	Susx	Jagdish Dilip Wagh	Artificial Intelligence And Data Science
2.	Susx	Rohit Krishna Chavan	Computer Science And Engineering
3.	Susx	Ibrahim Abdul Razzak Shaikh	Computer Science And Engineering
4.	Susx	Ayush Sharban Thakur	Artificial Intelligence And Machine Learning Engineering
5.	Bolts & Bots	Hrishikesh Nilesh Dhume	Electronics And Communication
6.	Bolts & Bots	Anmol Katiyar	Electronics And Communication
7.	Bolts & Bots	Soumil Chawan	Electronics And Communication
8.	Bolts & Bots	Archit Langde	Electronics And Communication
9.	Theinnovationarc	Sanjana Jagtap	Computer Science And Engineering
10.	Theinnovationarc	Sejal Gupta	Cyber Security
11.	Theinnovationarc	Prince Gupta	Artificial Intelligence And Machine Learning Engineering
12.	Theinnovationarc	Divyansh Pandey	Electrical Engineering
13.	Code Red	Aayush Manoj Pataskar	Electronics And Communication
14.	Code Red	Anvit Mulik	Electronics And Communication
15.	Code Red	Trushna Vilas Hadawale	Electronics And Communication
16.	Code Red	Disha Vora	Electronics And Communication
17.	Ohm My God	Harsh Kumar	Electronics And Communication
18.	Ohm My God	Shravan Tupe	Electronics And Communication
19.	Ohm My God	Vedant Irawati	Electronics And Communication
20.	Ohm My God	Tanisha Kothari	Electronics And Communication
21.	Wavex	Harsh Dhamane	Electronics And Communication
22.	Wavex	Hemanshu Bharambe	Electronics And Communication
23.	Wavex	Anushka Daple	Electronics And Communication
24.	Wavex	Ananya Pawar	Electronics And Communication

25.	Iotians	Om Gupta	Computer Science And Engineering
26.	Iotians	Arushi Shukla	Computer Science And Engineering
27.	Iotians	Rudra Raorane	Computer Science And Engineering
28.	Iotians	Nivedhithaa Selvakumar	Computer Science And Engineering
29.	Technojunkies	Shreya Manoj	Electronics And Communication
30.	Technojunkies	Kshyanika Behera	Electronics And Communication
31.	Technojunkies	Harini Mekala	Electronics And Communication
32.	Technojunkies	Pallavi Gali	Electronics And Communication
33.	Team Trishul	Fanindra Vaishnav	Computer Science And Engineering
34.	Team Trishul	Sparsh Singh	Electronics And Communication
35.	Team Trishul	Shubhank Kela	Electronics And Communication
36.	Hivemind	Sanmeet Dey	Robotics And Automation Engineering
37.	Hivemind	Anurag Rai	Computer Science And Engineering
38.	Hivemind	Mohammad Haris Khan	Robotics And Automation Engineering
39.	Techtribe	Harsh Rajesh Sankhe	Computer Science And Engineering
40.	Techtribe	Sneha Kaidan	Mechanical Engineering
41.	Neurovent	Ayush Brikeshkumar Pandey	Electronics And Communication
42.	Neurovent	Shivam Singh	Electronics And Communication
43.	Neurovent	Pradip Kumar	Electronics And Communication
44.	Neurovent	Taha Sayyad	Electronics And Communication
45.	Krishi Mitra	Satyam Chhatthu Mali	Electrical Engineering
46.	Krishi Mitra	Mohammad Ayaz Khan	Electrical Engineering
47.	Krishi Mitra	Siddhant Sachin Kadam	Electronics And Communication
48.	Krishi Mitra	Mandar Khadsan	Electrical Engineering
49.	Hackronauts	Abdul Qadeer Mujahid	Information Technology
50.	Hackronauts	Areeb Patel	Computer Science And Engineering
51.	Hackronauts	Hassaan Shaikh	Computer Science And Engineering
52.	Hackronauts	Haris Dhukka	Computer Science And Engineering
53.	Mistreated Devs	Karanjawane Parth	Electronics And Communication
54.	Mistreated Devs	Manav Kothari	Electronics And Communication
55.	Mistreated Devs	Tanish Nagarkar	Electronics And Communication
56.	Mistreated Devs	Yaswanth	Electronics And Communication
57.	Piston Power	Om Sanap	Mechanical Engineering
58.	Piston Power	Tanish Kunder	Mechanical Engineering
59.	Piston Power	Mustafa Pathan	Mechanical Engineering

60.	Piston Power	Aditya Parab	Mechanical Engineering
61.	Roboshake	Jaydatt Madan Sawant	Mechanical Engineering
62.	Roboshake	Ashutosh Mishra	Mechanical Engineering
63.	Roboshake	Kartik Nyalapelli	Electronics And Communication
64.	Voltmesh	Yashvi Jain	Electronics And Communication
65.	Voltmesh	Anu Saha	Electronics And Communication
66.	Voltmesh	Netra Hande	Electronics And Communication
67.	Voltmesh	Vipaak Gaikwad	Electronics And Communication
68.	Rjmask	Mrunmai Paktekar	Electronics And Communication
69.	Rjmask	Blossom Mascarenhas	Electronics And Communication
70.	Rjmask	Raman Sharma	Electronics And Communication
71.	Powerpuff	Neel Pujari	Electronics And Communication
72.	Powerpuff	Singh Priyanshukumar Ramasare	Electronics And Communication
73.	Powerpuff	Nisha Singh	Electronics And Communication
74.	Powerpuff	Onkar Lambade	Electronics And Communication
75.	3rd WHEEL	Jaeden Dsouza	Mechanical Engineering
76.	3rd WHEEL	Sumeet Gupta	Mechanical Engineering
77.	3rd WHEEL	Atharva Chache	Mechanical Engineering
78.	Smartdevs	Niraj Vikas Mahajan	Electronics And Communication
79.	Smartdevs	ADNAN QURESHI	Electronics And Communication
80.	Smartdevs	Aakib Patel	Electronics And Communication
81.	Smartdevs	Fayyaz Shaikh	Electronics And Communication
82.	Team404	Sahil Shingre	Electronics And Communication
83.	Team404	Sumit Desai	Electronics And Communication
84.	Team404	Aditya Thakur	Electronics And Communication
85.	Team404	KRITIKUMAR UMESH PAL	Electronics And Communication
86.	The Pathfinder	Sagnik Ulvekar	Mechanical Engineering
87.	The Pathfinder	Sahil Toskar	Mechanical Engineering
88.	Eterna Hackers	Kapil Labde	Electronics And Communication
89.	Eterna Hackers	Om Shashikant Patil	Computer Science And Engineering
90.	Eterna Hackers	Samiksha Gaonkar	Electronics And Communication
91.	Eterna Hackers	Umair Shaikh	Electronics And Communication

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Post of the student:
Reporting Head & Joint Reporting Head, IEEE-DBIT

Report Approved By: IEEE - DBIT

Name of the Faculty: Ms. Freda Carvalho

Post of the Faculty: IEEE-DBIT Student Branch Counsellor